International Institute of Medical Science & Technology Council

A Project Report On

"ED-FEED: AN AI-DRIVEN INTELLIGENT STUDENT FEEDBACK ANALYSIS AND INSIGHT GENERATION SYSTEM"

Submitted by Cohort – 21 (Data Science and Machine Learning)

Cohort Members:

1. Rajesh
2. Afra Fathima
3. Risul Dev
4. Aishwarya S Koti
5. Suhani C N
6. Yashwanth Ashok
7. Vishnu R Nair
8. Manav Mahesh
9. Rakshitha K
10. Disha Prakash
11. Rachana M L
12. Mandira P Gowda
13. Sowjanya R
14. Bhavana Suresh

TABLE OF CONTENTS

ABSTRACT

Educational institutions collect student feedback regularly, but the process of analyzing it remains largely manual, slow, and ineffective. Faculty coordinators and HODs spend hours going through hundreds of responses, and by the time any pattern is identified, the semester is already over. Negative trends go unnoticed until they reflect in poor results or student dropouts. The core challenge is that existing feedback systems only collect data and do not understand it. A student writing "the pace of the lectures is too fast and the assignments do not match what is taught in class" is expressing two distinct problems, but a star rating system reduces this to a 2 out of 5 with no further insight.

To address this, our team is building an AI powered real time student feedback analysis system. Student feedback is collected and fed into the system in real time as soon as it is submitted. The text is processed using NLP techniques including noise removal, tokenization and stopword removal to prepare it for analysis. A fine-tuned BERT model classifies each feedback as positive, negative or neutral and identifies the topic of concern such as teaching quality, course content, grading pattern or infrastructure. The system detects patterns over time and identifies if a particular subject or instructor is receiving a consistent surge of negative feedback. When a threshold is crossed, an automated alert is triggered and sent to the concerned coordinator or HOD (admin). An LLM integration using Gemini generates a plain English summary of what students are experiencing so that no manual reading is required. All insights are displayed on a live dashboard showing sentiment trends, topic wise breakdowns and time-based patterns in real time.

CHAPTER 1:

INTRODUCTION

In the evolving world of education, student feedback has become a primary indicator of the effectiveness of teaching, the quality of courses and the overall performance of institutions. It offers an immediate view into the student experience, allowing educators and administrators to refine pedagogical approaches and improve learning outcomes. However, the rapid digitisation of education has led to a significant increase in the amount of feedback obtained via online platforms. Manual and descriptive, traditional methods of analysing this feedback are no longer enough; they are time-consuming and inconsistent, and they are limited in their ability to reveal deeper patterns and sentiments within unstructured text. To bridge this gap, we propose ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System as a scalable and efficient solution. The system uses advanced Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to automate the extraction, classification and interpretation of student feedback. ED-FEED converts unstructured text data into structured insights, enabling educational institutions to make data-driven decisions to improve teaching quality and student satisfaction.

The system incorporates a hybrid sentiment analysis framework that seeks to provide contextual understanding and sequence-based learning. It uses a two-stage classification strategy that combines the reasoning capabilities of a Large Language Model (LLM) with the temporal modelling strengths of a Bidirectional Long Short-Term Memory (Bi-LSTM) network.

The architecture allows for accurate classification of the feedback into three sentiment categories, Positive, Negative, and Neutral, and successfully captures subtle expressions and contextual dependencies in student responses.

It culminates in a deployable inference pipeline that can process feedback at scale in real time. This enables institutions to remain in continuous contact with student opinions, to anticipate future problems and to make tangible improvements to teaching methods. ED-FEED helps to build more responsive, adaptive and student-centric educational environments by changing the analysis of feedback from a reactive task into a proactive, intelligence-driven process.

CHAPTER 2:

LITERATURE REVIEW

2.1 Review of Papers

2.1.1 Sentiment analysis and opinion mining on educational data: A survey

AUTHORS: Thanveer Shaik [a,*] , Xiaohui Tao [a] , Christopher Dann [b] , Haoran Xie [c] , Yan Li [a] , Linda Galligan [a]

1. Purpose Identified: To review sentiment analysis and opinion mining techniques across four levels: document, sentence, entity, and aspect in educational settings, covering annotation methods and AI models applied to course reviews, social media, and academic datasets.

2. Methodology Used: Systematic survey of 150+ studies using datasets from Coursera, Twitter, SFU corpus, and Amazon education reviews. Models reviewed include SVM, Random Forest, Naïve Bayes, XGBoost, CNN, LSTM, BiLSTM, BERT, RoBERTa, and transformer ensembles.

3. Results & Accuracy: Random Forest achieved F1 scores of 98.01% and 99.43% on aspect identification. Twitter education feedback reached 96.99% accuracy (aspects) and 88.72% (emotions). Coursera ABSA yielded an F1 of 86.13%.

4. Limitations & Future Scope: No new model proposed. Annotation tools are not adapted to educational language. Multilingual SA and real-time feedback pipelines remain unexplored in the literature.

5. Key Gaps Identified: Education-specific annotation tools and lexicons are absent. Real-time feedback pipelines and multilingual sentiment analysis in academic contexts are critical unaddressed needs.

2.1.2 Sentiment Analysis of Students' Feedback with NLP and Deep Learning: A Systematic Mapping Study

AUTHORS: Zenun Kastrati [1,*] , Fisnik Dalipi [1] , Ali Shariq Imran [2] , Krenare Pireva Nuci [3] and Mudasir Ahmad Wani [2]

1. Purpose Identified: To systematically map NLP, deep learning, and machine learning research (2015–2020) applied to student feedback on e-learning platforms, using the PRISMA framework across six major academic databases.

2. Methodology Used: PRISMA-based mapping of 92 primary studies distilled from 612 results across IEEE Xplore, ACM, Springer, Elsevier, Scopus, and Web of Science. Techniques mapped include NB, SVM, Decision Tree, Logistic Regression, LSTM, CNN, GRU, BERT, and hybrid combinations.

3. Results & Accuracy: Best reported accuracy across mapped studies: 98.29% with F1 of 96%. Typical applied study performance ranged from F1 = 83–88%, precision ~90%, and recall ~78%.

4. Limitations & Future Scope: No model or experiment conducted. No standardised datasets exist across studies. Emotional detection is significantly underdeveloped compared to polarity classification.

5. Key Gaps Identified: Standardised shared datasets for educational SA are absent. Emotional detection and real-time deployment have not been studied and both are identified as priority research directions.

2.1.3 Sentiment Analysis of Students' Feedback in MOOCs: A Systematic Literature Review

AUTHORS: Fisnik Dalipi [1 *], Katerina Zdravkova[2] and Fredrik Ahlgren[1]

1. Purpose Identified: To review how SA techniques have been applied to student feedback on MOOC platforms like Coursera, edX and Udemy and examining methods used to extract insights from course reviews and learner comments.

2. Methodology Used: PRISMA-based SLR restricted to English-language studies. Techniques reviewed include SVM (dominant), Decision Trees, CNN, LSTM, BERT, and LDA for topic modelling on platform-specific corpora.

3. Results & Accuracy: SVM consistently achieved F1 > 80%. Coursera ABSA reached F1 = 86.13%. LDA-based negative topic modelling achieved a precision of 85.71% for identifying negative feedback themes.

4. Limitations & Future Scope: English-only inclusion criteria limit global applicability. No cross-platform comparison conducted. No real-time feedback analysis system was found across all reviewed works.

5. Key Gaps Identified: Multilingual MOOC sentiment analysis is unexplored. Cross-platform benchmarking and longitudinal learner behaviour analysis are absent and real-time feedback loops in MOOC platforms remain unbuilt.

2.1.4 Students feedback analysis model using deep learning-based method and linguistic knowledge for intelligent educational systems

AUTHORS: Asad Abdi1[2], Gayane Sedrakyan[2,3] , Bernard Veldkamp[3], Jos van Hillegersberg[2] , Stéphanie M. van den Berg[3]

1. Purpose Identified: To propose DTLP- a novel BiLSTM model enriched with linguistic knowledge (POS tagging, negation detection, LDA) for fine-grained aspect-based sentiment analysis of student feedback in intelligent educational systems.

2. Methodology Used: Custom manually annotated student survey dataset covering course quality, teaching, and assignments. DTLP evaluated against CNN, SVM, Random Forest, and BERT baselines using accuracy and F1 as primary metrics.

3. Results & Accuracy: DTLP achieved 88.78% accuracy and F1 = 83.27%, outperforming CNN (71.43%) and matching SVM (86.01%), with the linguistic features providing measurable gains over pure neural baselines.

4. Limitations & Future Scope: Negation and discourse markers remain problematic for classification. Dataset is single-institution and domain-specific, limiting cross-institutional generalisation.

5. Key Gaps Identified: Larger and more diverse training corpora are needed. Transfer learning across institutions and improved negation handling are identified as priority future directions alongside real-time monitoring.

2.1.5 A systematic review of aspect-based sentiment analysis: domains, methods, and trends

AUTHORS: Yan Cathy Hua[1] , Paul Denny[1] , Jörg Wicker[1] , Katerina Taskova[1]

1. Purpose Identified: To present the largest SLR of ABSA research to date the tracking 20 years of trends, datasets, and solution paradigms from classical ML through transformer models to LLMs across multiple application domains.

2. Methodology Used: PDF-mining filter applied to 8,550 results, yielding 727 studies. Full method trajectory reviewed:

SVM → LSTM/BiLSTM → BERT/RoBERTa/XLNet → GPT-3.5/4, Llama-2, Flan-T5, DeBERTa, and PaLM.

3. Results & Accuracy: Fine-tuned RoBERTa outperformed GPT-3.5 and GPT-4 by ~20% on aspect extraction and sentiment classification. GPT-3.5 exceeded GPT-4 on opinion extraction by ~15% in zero-shot settings.

4. Limitations & Future Scope: Heavy bias towards SemEval benchmarks (restaurant and laptop domains). Education domain significantly underrepresented. LLM benchmarking protocols are inconsistent across studies.

5. Key Gaps Identified: ABSA must be diversified into education and healthcare domains. Multilingual ABSA datasets and standardised LLM benchmarking protocols are urgently needed.

2.1.6 Sentiment analysis for formative assessment in higher education: a systematic literature review

AUTHORS: Carme Grimalt-Álvaro1 , Mireia Usart

1. Purpose Identified: To review sentiment analysis applied specifically to formative assessment in higher education (2010–2022), covering techniques, visualisation tools, and pedagogical frameworks across discussion forums, evaluation forms, and peer/self-assessment data.

2. Methodology Used: PRISMA SLR covering HE discussion forums, teacher evaluation forms, and peer/self-assessment comments. Tools reviewed include SVM, VADER, SentiStrength, BERT, and hybrid rule-based and ML approaches.

3. Results & Accuracy: No single aggregate accuracy reported (review paper). Models incorporating emotional features consistently outperformed polarity-only approaches. Human rater involvement improved annotation validity and downstream classifier performance.

4. Limitations & Future Scope: Scarcity of student messages limits training data availability. SA tools are not adapted to academic language. Teacher-facing visualisation and dashboard tools are entirely absent from reviewed works.

5. Key Gaps Identified: Emotional state modelling and teacher dashboard development are unaddressed. Peer and self-assessment SA and student co-rater involvement are identified as priority future directions.

2.1.7 Elevating educational insights: sentiment analysis of faculty feedback using advanced machine learning models

AUTHORS: Sudhindra B. Deshpande[1] , Kiran K. Tangod[1] , Sowmyashree H. Srinivasaiah[2] , Ahmad Aziz Alahmadi[3] , Mamdooh Alwetaishi[4] , Goh Kah Ong Michael[5*] and Silambarasan Rajendran [6,7]

1. Purpose Identified: To apply and compare multiple classical machine learning classifiers on faculty feedback data student evaluations of teaching staff to identify the best-performing model for educational sentiment classification.

2. Methodology Used: Faculty evaluation forms preprocessed using NLP and POS tagging. Six classifiers evaluated: Random Forest, SVM, Naïve Bayes, Decision Tree, MLP, and SGD. BERT and RoBERTa proposed as future candidates.

3. Results & Accuracy: Random Forest achieved the best performance: 91% accuracy, 94% precision, 85% recall, and F1 = 89%. MLP and SGD underperformed significantly, confirming ensemble methods as superior for educational text.

4. Limitations & Future Scope: Traditional ML lacks contextual and semantic depth. Institution-specific data limits generalisability. MLP and SGD are not viable for this task without architectural improvements.

5. Key Gaps Identified: Fine-tuning BERT, RoBERTa, or EduBERT is the clear recommended next step. Severity detection and contextual sentiment categorisation across feedback dimensions remain unimplemented.

2.1.8 Formative Feedback on Student-Authored Summaries in Intelligent Textbooks Using Large Language Models

AUTHORS: Wesley Morris[1] , Scott Crossley[1] , Langdon Holmes[1] , Chaohua Ou[2] , Mihai Dascalu[3] , Danielle McNamara[4]

1. Purpose Identified: To fine-tune RoBERTa and Longformer within the iTELL intelligent textbook platform to automatically score student-written summaries across content and wording dimensions, delivering real-time formative feedback.

2. Methodology Used: 4,690 summaries from 3 HE institutions (2018–2021) across 101 source texts. Train/val/test split of 3,285/703/702 plus an out-of-domain Prolific set. PCA used to model content and wording quality separately.

3. Results & Accuracy: Content model: Pearson r = 0.78–0.81 vs human raters; PCA captured 82% variance. Wording model: 70% variance. Content model was robust out-of-domain; wording model degraded significantly outside training institutions.

4. Limitations & Future Scope: Wording model not generalisable out-of-domain. No XAI mechanism that is, scoring decisions are unexplainable to educators and students. System scope limited to textbook summarisation tasks.

5. Key Gaps Identified: Explainable AI for scoring decisions is the primary open problem. Expansion to broader writing and assessment contexts and curriculum redesign based on model metrics are identified as future work.

2.2 MARKET RESEARCH

The market for AI-powered student feedback analysis sits at the intersection of three high-growth sectors: the global EdTech market, the broader AI-in-education segment, and the specialized student feedback software niche. Understanding the size, dynamics, and competitive gaps in these markets is essential for positioning the present system and articulating its differentiated value proposition.

2.2.1 Market Size & Opportunity

The addressable market for an AI-driven feedback analysis solution is large and growing rapidly across all relevant segments. The following table summarizes key market data for 2024 with projections to 2027.

Market Segment		Size (2024)	Projected (2027)	CAGR
Global EdTech Market		$220 Billion	$400 Billion	18%
Student Feedback Software		$1.8 Billion	$2.5 Billion	12%
AI in Education		$5 Billion	$20 Billion	40%
India EdTech Market		$5.5 Billion	$10 Billion	25%

The AI-in-education segment stands out with a projected compound annual growth rate of 40% among the highest of any technology vertical reflecting the rapid adoption of intelligent tutoring systems, automated assessment tools, and adaptive learning platforms. The India EdTech market's 25% CAGR is particularly relevant given the density of engineering colleges, coaching institutes, and rapidly scaling online education platforms that constitute the primary target customer base for the proposed system.

2.2.2 Existing Solutions: General Feedback & Survey Platforms

The first tier of competitive landscape consists of general-purpose survey and experience management platforms that include basic feedback collection but lack the NLP depth and educational context required for meaningful student feedback analysis.

Company	Product	What It Does	Key Gap
Qualtrics	**XM Platform**	**Survey + basic sentiment analysis**	**Enterprise pricing ($15K+/yr), not education-specific**

SurveyMonkey	**Feedback Tool**	**Survey creation + basic reports**	**No AI, no real-time analysis, no NLP**
Medallia	**Experience Cloud**	**Customer feedback analysis**	**Corporate-focused, not designed for education**
Typeform	**Survey Tool**	**Beautiful forms + basic analytics**	**No sentiment analysis, no pattern detection**
Google Forms	**Free Survey**	**Data collection only**	**Zero analysis capability**

These platforms are built around the general enterprise feedback management use case. Their pricing models, particularly Qualtrics at $15,000+ per year, make them inaccessible to small and medium-sized educational institutions, and their NLP capabilities, where they exist at all, are trained on consumer and corporate feedback rather than academic discourse. None of these platforms can distinguish between feedback directed at a teacher, a course, or an institution's infrastructure.

2.2.3 Existing Solutions: Education-Specific Feedback Tools

The second tier of competitors consists of tools designed specifically for educational settings, primarily around structured course evaluation and reporting. While more contextually appropriate than general survey platforms, these tools rely on structured responses and do not employ NLP or AI to analyze free-text comments.

Company	**Product**	**What It Does**	**Key Gap**
Watermark	Course Evaluation	Student feedback collection + reports	Manual analysis, no AI, no real-time
EvaluationKIT	Course Evaluation	Structured reports + dashboards	No NLP, no free-text analysis
Campus Labs	Student Success	Engagement tracking	Not feedback-focused, no NLP
Explorance Blue	Academic Feedback	Structured feedback only	Cannot process free-text comments with AI

These education-specific tools serve their intended function of collecting and reporting structured evaluation data but are fundamentally limited by their reliance on quantitative ratings and Likert-scale responses. The rich qualitative information present in free-text student comments including nuanced critiques, emotional expressions, aspect-specific observations remain unprocessed and therefore unactionable. The absence of AI or NLP means that any insight beyond aggregate ratings must be generated through manual review, which is neither scalable nor timely.

2.2.4 Existing Solutions: AI-Powered Feedback Analytics

The third tier comprises AI-powered text analytics platforms that do apply NLP and sentiment analysis to feedback data. However, these tools are designed for customer experience management in commercial contexts and not for education and hence carry enterprise pricing that excludes smaller institutions.

Company	Product	What It Does	Key Gap
Thematic	Feedback Analysis	NLP-based theme extraction	Not education-specific, $500+/month
Kapiche	Text Analytics	Sentiment + theme detection	Enterprise pricing, generic use case
Chattermill	Unified Feedback AI	AI-powered customer insights	Designed for customer service, not education
MonkeyLearn	Text Analysis API	Sentiment + classification	Requires heavy customization, no dashboard

While these platforms demonstrate that AI-driven feedback analysis is technically feasible and commercially viable, their models are trained on customer service and product review data domains that differ substantially from academic feedback in vocabulary, structure, and analytical requirements. None of these tools support education-specific aspect detection (distinguishing feedback about teachers from feedback about course content), nor do they provide the difficulty-sentiment correlation that is central to the present project's design.

2.2.5 Gap Analysis: What the Market Is Missing

A systematic examination of all existing solutions across the three tiers above reveals five critical and consistently unaddressed gaps in the market. Each gap is supported by findings from the literature survey, confirming that these are not merely commercial opportunities but scientifically validated unmet needs.

Gap 1: Real-Time Analysis

All existing tools, whether general survey platforms, education-specific evaluation tools, or commercial AI analytics platforms process feedback in batches: weekly or monthly summary reports generated after data has accumulated. No tool provides instant sentiment detection as feedback is submitted, and no live dashboard updates or real-time alert mechanisms exist in the current market. Research across the literature survey consistently highlights the absence of real-time feedback pipelines as a critical unaddressed need (Shaik et al., 2023; Kastrati et al., 2021; Dalipi et al., 2022), confirming that this gap reflects a genuine technological frontier rather than a deliberate design choice.

Gap 2: Education-Specific NLP

Generic NLP models, whether deployed in commercial tools or trained on consumer review datasets, systematically misclassify academic language and context. No existing tool differentiates between feedback directed at a teacher's pedagogical approach, the inherent difficulty of course content, and complaints about physical or digital infrastructure. Annotation tools have not been adapted to educational language, and multilingual sentiment analysis in academic contexts remains largely unexplored (Shaik et al., 2023; Grimalt-Álvaro & Usart, 2024). The dominant ABSA benchmarks remain biased towards restaurant and laptop reviews, leaving the education domain significantly underrepresented in the models that commercial tools use (Hua et al., 2024).

Gap 3: Difficulty-Sentiment Intelligence

No existing tool combines difficulty index with sentiment polarity to derive root cause attribution that is, a capability with substantial diagnostic value. The critical insight that a low rating paired with high reported difficulty points to a course design problem, while the same low rating paired with low difficulty indicates a teaching effectiveness issue, is absent from all current solutions. Fine-grained aspect-based sentiment analysis on student feedback has shown that negation, discourse markers, and domain-specific context significantly affect classification accuracy, requiring purpose-built models rather than generic tools (Abdi et al., 2023).

Gap 4: Affordability & Accessibility

Enterprise tools such as Qualtrics and Medallia carry annual license costs of $5,000 to $50,000, placing them beyond the reach of small colleges, coaching institutes, and early-stage EdTech startups, precisely the institutions that generate the most student feedback and have the least capacity to analyze it manually. A systematic mapping of 92 primary studies found no standardized datasets and no accessible shared tools for educational sentiment analysis, highlighting the lack of open, institution-ready solutions (Kastrati et al., 2021). There is no affordable, scalable option in the current market for small-to-medium educational institutions.

Gap 5: Actionable LLM-Powered Insights

Existing tools surface data in the form of charts, aggregate scores, and raw comment listings but do not generate recommendations. Educators must interpret raw analytical outputs themselves, a process that is time-consuming, requires statistical literacy, and is frequently deprioritized given competing institutional demands. No tool currently uses large language models to auto-generate human-readable summaries, contextualized interpretations, and actionable recommendations from feedback data. Recent research on automated formative feedback using fine-tuned transformer models demonstrates strong correlation with human raters, yet highlights the absence of explainability and actionable output for educators (Morris et al., 2025). The trajectory of ABSA research from SVM to BERT to LLMs confirms that the next frontier is deploying these models in real educational settings with interpretable, actionable outputs (Hua et al., 2024; Deshpande et al., 2025).

2.2.6 Competitive Advantage Comparison

The following table positions the proposed system against existing solution categories across eight key capability dimensions, illustrating how the identified market gaps translate into direct competitive differentiators.

Feature	Traditional Tools	Generic AI Tools	Our Solution
Real-Time Analysis	No	Partial	Yes, Live Pipeline
Education-Specific NLP	No	No	Yes, BERT Fine-tuned
Difficulty-Sentiment Matrix	No	No	Yes, Unique Feature
LLM Auto Summaries	No	Partial	Yes, Gemini Integration
Pattern & Trend Detection	Basic	Partial	Yes, Time-series
Alert System	No	No	Yes, Threshold-based
Affordable Pricing	No	No	Yes, Accessible
Interactive Dashboard	Basic	Basic	Yes, Real-time

Comparative studies on ML classifiers for faculty feedback evaluation show Random Forest achieving 91% accuracy, establishing a strong baseline that the proposed BERT-based architecture aims to surpass through richer contextual understanding (Deshpande et al., 2025). Further, fine-tuned RoBERTa has been shown to outperform GPT-3.5 and GPT-4 by approximately 20% on aspect extraction and sentiment classification tasks validating the use of fine-tuned transformer models over generic LLM deployments as the core of the proposed classification engine (Hua et al., 2024).

CHAPTER 3:

SYSTEM ANALYSIS

System analysis forms the foundation upon which the design and development of any software-intensive project is grounded. It encompasses a structured examination of the problem domain, the functional and non-functional requirements of the proposed solution, and the constraints within which the system must operate. For the AI-Based Student Feedback Analysis System, the system analysis phase involves a thorough understanding of how educational feedback is currently collected and processed, the limitations of existing approaches, and the specific capabilities that the proposed system must deliver. This chapter covers the objectives of the system, a description of the proposed system, and a detailed breakdown of both software and hardware requirements necessary for its implementation.

3.1 PROBLEM STATEMENT

Educational institutions routinely collect student feedback, but its analysis remains manual, time-consuming, and often delayed, limiting its effectiveness. Existing systems primarily capture data without interpreting it, reducing detailed student concerns into simplistic metrics like ratings, which fail to convey meaningful insights. As a result, critical issues and negative trends often go unnoticed until they impact academic outcomes. To address this gap, the proposed system leverages AI and NLP to analyse student feedback in real time, classify sentiments, identify key concerns, and detect emerging patterns. It provides automated alerts and concise summaries, enabling institutions to take timely, data-driven actions and improve overall educational quality.

3.2 OBJECTIVES

The primary objective of this project is to design, develop, and deploy an end-to-end artificial intelligence system capable of automatically analyzing student feedback with high accuracy and interpretability. Unlike conventional feedback analysis approaches that rely on manual review or simple keyword matching, the proposed system leverages state-of-the-art natural language processing techniques specifically a Bidirectional Long Short-Term Memory (Bi-LSTM) architecture combined with LLM-assisted data labelling to produce a robust, scalable, and privacy-compliant feedback intelligence pipeline. The following objectives define the scope and intent of the system:

- To build a robust three-class sentiment classifier distinguishing Positive, Negative, and Neutral feedback trained on real-world educational data sourced from platforms such as Coursera, RateMyProfessor, and institutional review corpora.
- To improve training data quality through LLM-assisted re-labelling, utilizing LLaMA 3.1 8B Instant via the Groq Cloud API with zero-shot classification prompts at temperature 0.0 to ensure deterministic and consistent ground-truth annotations across 35,000 feedback comments.
- To implement a privacy-compliant preprocessing pipeline that anonymizes personally identifiable information (PII) including names, email addresses, phone numbers, and student IDs using regular expression pattern matching and spaCy Named Entity Recognition, ensuring compliance with data protection standards before any model training or inference occurs.
- To achieve high classification performance across all three sentiment classes using a Bidirectional LSTM architecture that processes sequential text in both temporal directions, enabling the model to capture complex negation, qualification, and contextual dependencies that are characteristic of academic feedback language.
- To package the trained model into a reusable, low-latency inference pipeline suitable for real-time deployment, with inference times of under 500 milliseconds per comment on standard CPU hardware, and to expose predictions through a teacher-facing dashboard that visualizes sentiment distributions, trend lines, and flagged high-concern responses.

3.3 PROPOSED SYSTEM

The proposed system is an intelligent, end-to-end feedback analysis platform that automates the entire journey from raw student comment to actionable educator insight. At its core, the system addresses three fundamental shortcomings of current feedback processing practices in educational institutions: the absence of scalable automated analysis, the lack of data privacy safeguards in feedback handling, and the inability to extract fine-grained, aspect-level sentiment from unstructured free-text responses.

The system ingests student feedback from multiple collection points web forms, LMS-integrated portals, and standalone Google Forms and subjects each response to a structured eight-stage processing pipeline. In the first stage, raw text is anonymized to replace all PII with safe placeholder tokens, ensuring that downstream model training and inference operate on privacy-compliant data. The pre-processed text is then passed through TensorFlow's TextVectorization layer, which tokenises comments on whitespace, constructs a vocabulary of the 20,000 most frequent tokens, and pads or truncates each sequence to a uniform length of 100 tokens producing input tensors ready for neural network processing.

The classification engine is a Bidirectional LSTM network implemented in TensorFlow Keras, trained on 70% of the LLM-relabelled dataset with a 15% validation and 15% held-out test split, using stratification to maintain class proportions across all subsets. Training leverages the Adam optimiser with an initial learning rate of 0.001, categorical cross-entropy loss, and three callbacksEarly Stopping, Model Checkpointing, and ReduceLROnPlateau to prevent overfitting and recover the best-performing model weights. Distributed training across two NVIDIA T4 GPUs using TensorFlow's MirroredStrategy halves training time without requiring changes to the model architecture.

Once trained and validated, the saved model checkpoint powers a real-time inference endpoint that classifies new feedback comments within milliseconds. Predictions, sentiment trends, aspect-level breakdowns, and flagged items are surfaced through a web-based dashboard accessible to faculty and department administrators, enabling data-driven pedagogical decisions at both the individual course and institutional level.

CHAPTER 4:

REQUIREMENT ANALYSIS

Requirement analysis identifies the conditions that the system must satisfy both functional and non-functional requirements and the physical and computational resources needed to build, train, and deploy it. The requirements below are derived from the project objectives, the eight-stage processing architecture, and the operational constraints of the target deployment environment.

4.1 SOFTWARE REQUIREMENTS

The software stack for this project spans data engineering, machine learning, natural language processing, model deployment, and visualisation. All core components are open-source and cross-platform, enabling reproducibility across development, training, and production environments.

Component	Technology / Library	Purpose
Programming Language	Python	Core development language for all pipeline stages
Deep Learning Framework	TensorFlow / Keras	Bi-LSTM model definition, training, and inference
NLP Preprocessing	spaCy	Named Entity Recognition for PII anonymization
NLP Utilities	NLTK	Stopword lists, tokenization support
ML Utilities	scikit-learn	LabelEncoder, stratified train/val/test splitting
LLM Re-labelling	Groq Cloud API + LLaMA 3.1 8B	Zero-shot deterministic sentiment annotation
Data Handling	pandas / NumPy	Dataset manipulation, CSV I/O, array operations
Visualization	Matplotlib / Seaborn	Training curves, confusion matrices, evaluation plots
Dashboard	React/ Flask	Real-time web-based educator dashboard
GPU Training	TensorFlow MirroredStrategy	Synchronous distributed training across 2 T4 GPUs
Development Environment	Kaggle Notebooks	2x NVIDIA T4 GPU cloud environment
Version Control	Git / GitHub	Source code management and collaboration

4.2 HARDWARE REQUIREMENTS

The hardware requirements are specified at two levels: the development and training environment, which demands GPU-accelerated compute for model training on 35,000 labelled samples; and the inference and deployment environment, which requires only modest CPU resources given the low-latency (<500 ms) inference target.

Component	Training Environment	Deployment / Inference Environment
Processor (CPU)	Intel Xeon or AMD EPYC (multi-core)	Intel Core i5 / i7 or equivalent (4+ cores)
GPU	2× NVIDIA Tesla T4 (16 GB VRAM each)	Not required (CPU inference sufficient)
RAM	16 GB minimum, 32 GB recommended	8 GB minimum
Storage	50 GB SSD (dataset + checkpoints + logs)	10 GB SSD (model weights + application)
Network	High-bandwidth internet (Groq API calls)	Standard broadband (dashboard access)
Platform	Google Colab Pro+ / Cloud VM (GPU-enabled)	On-premises server or cloud VM (CPU tier)

4.3 FUNCTIONAL REQUIREMENTS

The system must fulfil the following core functional requirements to meet the project objectives:

1. Accept free-text student feedback as input from web forms or LMS-integrated portals and associate each entry with metadata such as course ID, instructor ID, anonymized student ID, and semester.
2. Anonymize all PII from raw feedback text using regex pattern matching and spaCy NER before any downstream processing or storage.
3. Vectorize preprocessed text into fixed-length integer sequences using a 20,000-token vocabulary and a sequence length of 100 tokens.
4. Classify each feedback comment into one of three sentiment classes: Positive, Negative, or Neutral, using the trained Bi-LSTM model with softmax probability output.
5. Support real-time inference with latency under 500 milliseconds per comment on CPU hardware.
6. Display sentiment distributions, trend analyses, aspect-level breakdowns, and flagged high-concern feedback on a real-time web dashboard accessible to faculty and administrators.

7. Cache all LLM re-labelling results to a CSV file to prevent redundant Groq API calls across pipeline runs.

4.4 NON-FUNCTIONAL REQUIREMENTS

Beyond functional correctness, the system must satisfy the following quality and operational constraints:

- Privacy & Compliance: All PII must be anonymized prior to model training and inference. No raw identifiable feedback may be stored or transmitted to external services.
- Scalability: The vectorization and inference pipeline must handle batch inputs without architectural changes, supporting institutional-scale volumes of feedback (thousands of comments per semester).
- Reproducibility: LLM re-labelling at temperature 0.0 and seeded data splits ensure fully deterministic pipeline behavior across runs.
- Reliability: Exponential backoff and retry logic in the Groq API client ensure graceful handling of transient API failures during the labelling stage.
- Maintainability: Modular pipeline stages like anonymization, vectorization, model inference, and dashboard are independently testable and replaceable.

CHAPTER 5:

METHODOLOGY

The methodology of this project follows a structured, multi-stage machine learning pipeline designed to address the specific challenges of educational sentiment classification namely, the need for high-quality labelled data, privacy-compliant preprocessing and a classification architecture capable of capturing the linguistic nuances of student feedback. The pipeline is divided into three core phases: LLM-assisted data re-labelling, text preprocessing and feature engineering, and Bidirectional LSTM model architecture and training. Each phase is described in full below.

5.1 LLM-ASSISTED RE-LABELLING

A foundational challenge in training any supervised sentiment classifier is the quality and consistency of ground-truth labels. Manual annotation is expensive, time-consuming, and susceptible to inter-annotator disagreement particularly for feedback text that uses hedged, sarcastic, or domain-specific language. To address this, the project employs a large language model to systematically re-label the entire dataset, correcting inconsistencies in any prior manual annotations and establishing a clean, deterministic ground truth.

Each of the 35,000 feedback comments was passed individually to LLaMA 3.1 8B Instant, Meta AI's open-source large language model accessed through the Groq Cloud API. A zero-shot prompt instructed the model to classify each comment into one of three sentiment categories (Positive, Negative, or Neutral) and respond with a single word only. The model configuration was set to a temperature of 0.0 to ensure fully deterministic, non-probabilistic output that is, the same comment will always receive the same label regardless of when the API call is made. The maximum response was limited to five tokens to prevent the model from producing explanatory text alongside the label.

To ensure pipeline robustness, the API client was equipped with retry logic using exponential backoff, gracefully handling transient network failures or rate-limit responses from the Groq endpoint. All returned labels were written incrementally to a CSV cache file, so that if the pipeline is interrupted and re-run, previously labelled comments are not re-submitted to the API thereby preserving both cost efficiency and result consistency. This LLM-assisted labelling strategy effectively transforms an unreliable or partially labelled dataset into a high-confidence, uniformly annotated corpus suitable for supervised deep learning.

5.2 DATA PREPROCESSING

Raw student feedback text cannot be fed directly into a neural network. The preprocessing stage transforms unstructured free-text comments into clean, anonymized, fixed-length numeric tensors through a sequence of well-defined operations, each designed to eliminate noise while preserving the semantic content relevant to sentiment classification.

Text anonymization is the first and most critical operation in the preprocessing pipeline. Regular expressions are applied to detect and replace sensitive patterns such as email addresses, phone numbers, URLs, student ID numbers, and numeric identifiers, with standardized placeholder tokens (EMAIL, PHONE, URL, ID, NUM respectively). This ensures that no personally identifiable information propagates into the feature engineering or model training stages, maintaining FERPA compliance and protecting student privacy throughout the system.

Following anonymizations, string sentiment labels (Positive, Negative, Neutral) are encoded as integers using scikit-learn's LabelEncoder, then converted to one-hot vectors to serve as targets for the categorical cross-entropy loss function used during training. The full dataset is then partitioned into training (70%), validation (15%), and test (15%) subsets using stratified splitting, which preserves the distribution of all three sentiment classes across each partition which is a critical step given that real-world educational feedback datasets are rarely perfectly balanced.

Text vectorization is handled by TensorFlow's built-in TextVectorization layer, which tokenizes each comment on whitespace, constructs a vocabulary of the 20,000 most frequent tokens from the training corpus, and maps each token to its integer index. Sequences shorter than 100 tokens are zero-padded, and those longer are truncated, producing uniformly shaped input tensors of dimension (batch_size, 100) that are directly consumable by the embedding layer of the Bi-LSTM model.

5.3 MODEL ARCHITECTURE

The classification model is a Bidirectional Long Short-Term Memory (Bi-LSTM) network implemented using TensorFlow Keras. The architectural choice is motivated by the nature of educational feedback text: student comments frequently employ negation ('not clear', 'barely engaging'), qualification ('somewhat helpful'), and discourse structures where meaning depends on words appearing both before and after the focal term. A standard unidirectional LSTM processes text from left to right only, which means it cannot access future context when encoding each position. A Bidirectional LSTM runs two LSTMs simultaneously: one reading the token sequence in the natural left-to-right order, and one reading it in reverse and concatenates their hidden states at each time step, giving every position full access to its surrounding context in both directions.

The six-layer architecture is designed to progressively transform a raw token sequence into a compact sentiment representation. The Embedding Layer maps each integer token ID to a trainable 64-dimensional continuous vector. Semantically related words converge towards nearby positions in this embedding space during backpropagation, allowing the model to generalize across synonyms and paraphrases without explicit feature engineering. A SpatialDropout1D layer with a dropout rate of 20% follows immediately, applying sequence-level regularization by randomly zeroing entire embedding vectors (rather than individual elements) during training. This sequence-level dropout is particularly well-suited to recurrent architectures because it disrupts feature co-adaptation across time steps more effectively than element-wise dropout.

The first Bidirectional LSTM layer contains 64 units in each direction (128 total), with return_sequences set to True, meaning it outputs a hidden state vector at every time step rather than only at the sequence endpoint. This allows the second Bi-LSTM layer, which contains 32 units per direction (64 total) and operates with return_sequences set to False, to process the full-sequence context produced by the first layer and compress it into a single 64-dimensional context vector representing the entire comment. This architectural bottleneck forces the model to distil the most sentiment-relevant information from the variable-length input into a fixed-size representation.

A Dense layer with 64 units and ReLU activation introduces non-linear decision boundaries, followed by an aggressive Dropout rate of 40% to prevent overfitting at the classification head. The final Output Dense layer contains 3 units, one per sentiment class with Softmax activation, which converts the raw output scores into a probability distribution summing to 1.0. The class with the highest predicted probability is selected as the model's prediction during inference.

CHAPTER 6:

SYSTEM DESIGN

The system architecture of the AI-Based Student Feedback Analysis System is structured as an eight-stage sequential processing pipeline, where each stage transforms the data output of the previous stage and passes it forward. The architecture is designed for modularity, each stage is independently replaceable and for compliance, with privacy safeguards applied at the earliest possible point in the pipeline. The diagram below illustrates the complete data flow from raw feedback ingestion to educator-facing dashboard output.

6.1 ARCHITECTURE OVERVIEW

The architecture diagram illustrates the complete processing pipeline of the system, organized as a vertical data-flow from the student feedback data sources at the top to the educator-facing dashboard output at the bottom. The pipeline is divided into eight discrete and sequentially dependent stages. Raw feedback enters from real-world dataset sources: RateMyProfessor instructor evaluations, Waterloo student reviews, Coursera course reviews and Exeter student evaluations providing linguistic and contextual diversity representative of authentic educational feedback. Each stage is described in detail below.

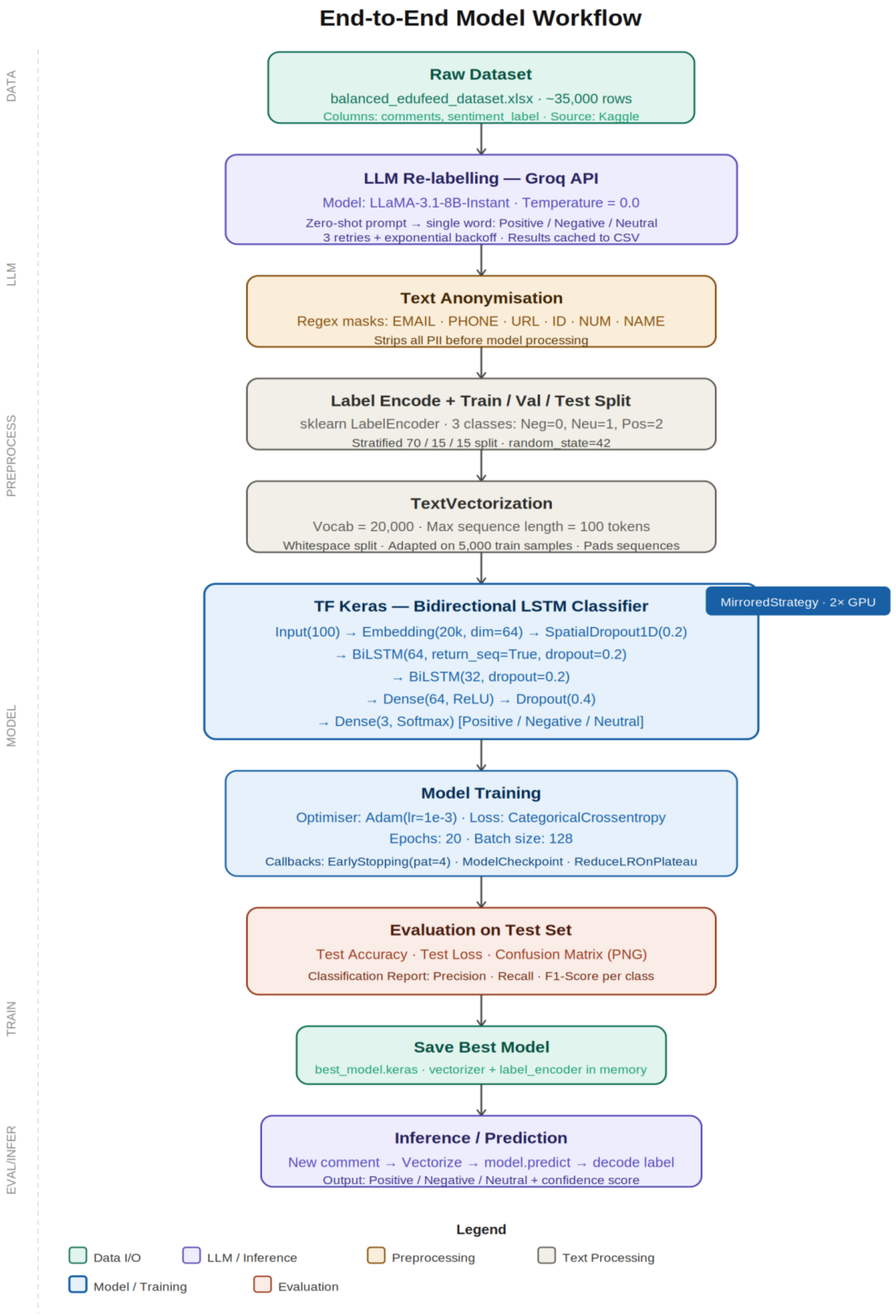
6.1.1 Data Collection and Ingestion

Student feedback is collected through web forms integrated with institutional Learning Management Systems or deployed as standalone Google Forms. Each feedback entry is timestamped and tagged with course ID, instructor ID, anonymized student ID, and semester identifier, ensuring that downstream analyses can be disaggregated by course, instructor, or time period. Ingested entries are stored in a staging database pending preprocessing.

6.1.2 Anonymization and Privacy Compliance

Raw feedback text is scanned for personally identifiable information using a dual-method approach: regular expression pattern matching identifies structured PII such as email addresses, phone numbers, URLs, and numeric identifiers, while spaCy's Named Entity Recognition module detects unstructured PII such as personal names and institutional affiliations. All detected PII is replaced with standardized placeholder tokens as NAME, EMAIL, PHONE, URL, ID, NUM before any data is written to a downstream processing stage. This anonymization step ensures compliance with the Family Educational Rights and Privacy Act

(FERPA) and equivalent data protection frameworks, and is applied unconditionally before model training or inference.



6.1.3 Text Preprocessing

Anonymized text undergoes standard NLP preprocessing to reduce noise and normalize vocabulary. Operations applied in sequence include: lowercasing all characters to eliminate case-based duplicates; whitespace and punctuation tokenization using spaCy; removal of standard English stop words using the NLTK stop-word list; and morphological stemming using the Porter Stemmer, which reduces inflected word forms to their common stem. Punctuation and special characters are normalized to remove non-informative symbols while preserving sentiment-bearing punctuation such as exclamation marks.

6.1.4 LLM-Assisted Deterministic Labelling

Each preprocessed comment is submitted to LLaMA 3.1 8B Instant via the Groq Cloud API using a zero-shot classification prompt: 'Classify this student feedback as Positive, Negative, or Neutral. Respond with one word only.' The model operates at temperature 0.0 for fully deterministic output, ensuring that the same comment always receives the same label across multiple pipeline runs. The returned single-word label becomes the ground truth for supervised training, effectively correcting inconsistencies inherited from any prior manual annotations. Results are cached to a CSV file to avoid redundant API calls.

6.1.5 Feature Engineering and Vectorization

Preprocessed and labelled comments are converted to fixed-length numeric sequences using TensorFlow's TextVectorization layer. The layer performs whitespace tokenization, builds a vocabulary from the 20,000 most frequent tokens in the training corpus, and maps each token to its vocabulary index. Sequences are padded with zeros or truncated to a uniform length of 100 tokens, producing input tensors of shape (batch_size, 100) that are directly consumable by the neural network's Embedding layer.

6.1.6 Bi-LSTM Model Training and Evaluation

The Bidirectional LSTM model is trained on 70% of the labelled dataset with a 15% validation split and evaluated on the remaining 15% held-out test set, using stratified splitting throughout to preserve class proportions. The full layer-by-layer architecture:

Embedding → SpatialDropout1D → BiLSTM (128 units) → BiLSTM (64 units) → Dense + Dropout → Softmax Output

6.1.7 Real-Time Inference and Classification

Once training is complete, the saved model checkpoint serves new, unseen feedback through a real-time inference pipeline. The inference path processes each raw comment through the identical TextVectorization layer (applying the same vocabulary and padding) before passing the resulting tensor through a forward pass of the Bi-LSTM network. The softmax output vector produces class probabilities; the class with the highest probability is selected via argmax as the model's prediction. End-to-end inference latency is consistently under 500 milliseconds per comment on standard CPU hardware, making the system suitable for real-time deployment in institutional settings.

6.1.8 Visualization and Teacher Dashboard

Sentiment predictions, aggregated trend analyses, aspect-level sentiment breakdowns, and flagged high-concern feedback items are visualized in a real-time web-based dashboard accessible to faculty and department administrators. The dashboard presents five key analytical views: an overall sentiment distribution pie chart showing the proportion of Positive, Negative, and Neutral feedback; a sentiment trend line chart tracking sentiment trajectory across the course duration; a horizontal bar chart showing aspect-level sentiment breakdown (teaching quality, assignment difficulty etc.); a ranked list of the top positive and top negative feedback excerpts; and a flagged items panel surfacing responses that exceed a configurable concern threshold and require immediate attention from the instructor or department head.

6.2 BI-LSTM LAYER ARCHITECTURE

The table below provides a concise technical summary of each layer in the Bidirectional LSTM model, including units, activations, and the functional role of each component in the overall classification pipeline.

Layer	Configuration	Output Shape	Role
Embedding	Vocab: 20,000 tokens | Dim: 64	(batch, 100, 64)	Maps integer token IDs to trainable 64-D semantic vectors
SpatialDropout1D	Rate: 20%	(batch, 100, 64)	Sequence-level regularization; zeros entire embedding vectors to prevent co-adaptation
Bidirectional LSTM 1	64 forward + 64 backward = 128 units | return_sequences=True	(batch, 100, 128)	Captures full bi-directional context at every sequence position; feeds full sequence to next layer
Bidirectional LSTM 2	32 forward + 32 backward = 64 units |	(batch, 64)	Compresses full-sequence context into a single 64-D

	return_sequences=False		vector; bottleneck representation
Dense + Dropout	64 units | ReLU | Dropout 40%	(batch, 64)	Non-linear transformation; aggressive regularization to prevent overfitting at classifier head
Output Dense	3 units | Softmax	(batch, 3)	Produces class probability distribution over Positive, Negative, Neutral

The layered design reflects a deliberate compression strategy: the first Bi-LSTM layer retains full sequence context by returning outputs at every time step (return_sequences=True), while the second Bi-LSTM layer collapses this into a single fixed-size summary vector (return_sequences=False). This two-stage recurrent structure followed by a non-linear dense classifier with heavy dropout balances expressive capacity against overfitting risk, which is a well-established design pattern for text classification on moderately sized labelled corpora.

CHAPTER 7:

TESTING

Testing is an integral part of the software development lifecycle, and for a machine learning system like the AI-Based Student Feedback Analysis System, it carries additional significance. Unlike conventional software where correctness can often be determined through boolean pass/fail conditions, an intelligent system must be verified not only for functional accuracy but also for model reliability, pipeline integrity, data handling correctness, and end-to-end behavioural consistency. This chapter documents the testing strategy adopted for the project, covering unit testing of individual pipeline components, integration testing of inter-module data flows, and system-level testing of the complete feedback processing pipeline.

7.1 UNIT TESTING

Unit testing focuses on verifying the correctness of individual, isolated components within the system. Each module in the pipeline was tested independently to ensure it performed its designated function accurately before being integrated with the rest of the system. This approach enabled early detection of defects and simplified debugging by narrowing the scope of any failure to a specific component.

7.1.1 Anonymisation Module

The anonymisation module is responsible for detecting and replacing personally identifiable information in raw student feedback before any downstream processing occurs. Unit tests were written to verify that all supported PII categories were correctly identified and substituted with their corresponding placeholder tokens.

Test cases covered a range of realistic input patterns, including standard email address formats, international and domestic phone number formats, HTTP and HTTPS URLs, numeric student ID strings of varying lengths, and standalone numeric values embedded in feedback text. Each test passed a crafted input string to the anonymisation function and asserted that the output contained the correct placeholder token (EMAIL, PHONE, URL, ID, or NUM) in place of the original sensitive value, and that no other characters in the string were unintentionally modified. Additional edge cases tested included strings containing multiple PII types simultaneously, feedback where numeric values appeared as part of valid sentiment words (such as scores or ratings), and comments that contained no PII at all, confirming that clean inputs passed through unaltered. All unit tests for this module passed successfully, validating that the anonymisation layer provides reliable privacy protection before any data reaches the machine learning pipeline.

7.1.2 Text Vectorisation Module

The text vectorisation module converts pre-processed string input into fixed-length integer sequences suitable for consumption by the neural network. Unit tests verified three core behaviours of the TensorFlow TextVectorization layer: vocabulary construction, sequence tokenisation, and padding and truncation logic.

Vocabulary tests confirmed that the layer correctly built a vocabulary from the training corpus limited to the 20,000 most frequent tokens, and that out-of-vocabulary tokens were assigned the reserved unknown token index. Tokenisation tests verified that whitespace-split tokens were mapped to their correct vocabulary indices consistently across multiple calls. Padding and truncation tests submitted sequences of lengths below, at, and above the target length of 100 tokens, asserting that all outputs were exactly 100 tokens in length regardless of input length. Short sequences were verified to be zero-padded from the right, and long sequences to be truncated at the 100-token boundary. All test cases produced the expected output shapes and values.

7.1.3 Label Encoding Module

The label encoding module maps raw string sentiment labels returned by the LLM re-labelling process to integer class indices, and subsequently converts them to one-hot vectors for use with the categorical cross-entropy loss function. Unit tests verified that scikit-learn's LabelEncoder assigned consistent indices to each class label across encoding and inverse-encoding operations, and that the resulting one-hot vectors had the correct shape and contained exactly one active element per sample.

Tests also confirmed that the stratified train/validation/test split preserved the class distribution within acceptable margins across all three subsets. For a balanced dataset where each of the three sentiment classes constitutes approximately one third of the total, the split was verified to maintain proportions within plus or minus two percentage points of the expected distribution in every partition.

7.1.4 LLM Re-Labelling Client

The Groq API client module was unit tested to verify its error handling, retry logic, and caching behaviour. Tests simulated transient API failures and confirmed that the exponential backoff mechanism retried the request the expected number of times before either succeeding or raising a logged exception. The CSV caching layer was tested by pre-populating the cache file with a set of labelled comments and verifying that subsequent pipeline runs retrieved cached labels without making API calls for those comments. A separate test confirmed that new, uncached comments were correctly submitted to the API and their returned labels appended to the cache file.

7.2 INTEGRATION TESTING

Integration testing examined the correctness of data flow and interface compatibility between connected modules. Rather than testing components in isolation, integration tests exercised module pairs and small chains of modules to verify that outputs produced by one stage were correctly consumed by the next, and that no data was lost, incorrectly transformed, or improperly formatted at module boundaries.

7.2.1 Anonymisation to Preprocessing Integration

The first integration test connected the anonymisation module to the downstream text preprocessing steps, which include lowercasing, tokenisation, stop word removal, and stemming. The test verified that placeholder tokens introduced by the anonymisation stage (such as EMAIL and PHONE) were preserved intact through the preprocessing steps and not inadvertently stripped by the stop word filter or stemmer. Since placeholder tokens do not appear in standard English stop word lists and the Porter Stemmer treats them as normal words, they were confirmed to pass through the preprocessing chain without modification, arriving at the vectorisation stage with their intended form.

7.2.2 Preprocessing to Vectorisation Integration

This integration test confirmed that the string output produced by the text preprocessing stage was correctly accepted as input by the TensorFlow TextVectorization layer. The test exercised batches of pre-processed comments and verified that the vectorisation layer produced output tensors of the expected shape and data type. Particular attention was paid to the handling of empty strings, which could arise if a comment consisted entirely of stop words and was reduced to an empty sequence after preprocessing. The test confirmed that empty inputs were mapped to a zero-padded sequence of length 100 rather than causing a runtime error.

7.2.3 Vectorisation to Model Inference Integration

The integration between the vectorisation layer and the Bidirectional LSTM model was tested by passing a batch of vectorised sequences through the full model forward pass and verifying that the softmax output layer produced valid probability distributions. Each output vector was confirmed to contain exactly three values summing to 1.0 within floating-point tolerance, with all individual probabilities falling in the range of 0 to 1. The argmax of each output vector was also verified to correspond to one of the three valid class indices, confirming that the model produced actionable predictions for every input in the batch.

7.2.4 Model Output to Dashboard Integration

The final integration test verified the flow of model predictions into the visualisation and dashboard layer. Predicted class labels, confidence scores, and associated metadata (course ID, instructor ID, semester) were passed to the dashboard data pipeline and verified to be correctly aggregated into sentiment distribution counts, time-series trend values, and flagged item lists. The test confirmed that the dashboard rendered the correct sentiment proportions for a known test batch, that trend charts updated correctly when predictions from multiple time periods were supplied, and that the flagging threshold correctly identified comments whose Negative class probability exceeded the configurable alert level.

7.3 SYSTEM TESTING

System testing evaluated the behaviour of the complete, end-to-end feedback analysis pipeline operating as a unified system. Unlike unit and integration tests, which focus on individual components or module pairs, system testing submits realistic inputs at the pipeline entry point and validates the outputs at the final stage, treating the intermediate processing as a black box. This level of testing is the closest approximation to real-world operational conditions and provides the most direct evidence of the system's fitness for deployment.

7.3.1 End-to-End Pipeline Test

A representative sample of 500 student feedback comments was assembled from the held-out test set to serve as the system test corpus. These comments spanned all three sentiment classes, included a variety of sentence structures and vocabulary patterns, and contained intentionally embedded PII to verify the full effect of the anonymisation stage. Each comment was submitted to the pipeline entry point without any preprocessing, and the system was allowed to execute all eight stages autonomously.

The system test verified that all PII was correctly anonymised in the first stage and was absent from all subsequent pipeline stages and stored outputs. It confirmed that pre-processed, vectorised tensors were correctly shaped before entering the model. The model produced a valid three-class probability distribution for every comment, and the predicted class labels were written to the output store with the correct metadata associations. The dashboard reflected the updated predictions in real time after the batch was processed. No runtime errors, data type mismatches, or silent failures were observed during any run of the end-to-end test.

7.3.2 Privacy and Compliance Test

A dedicated system test was conducted to verify that no personally identifiable information propagated beyond the anonymisation stage into any stored artefact, log file, model weight, or dashboard output. A set of 50 feedback comments containing known PII patterns was passed through the full pipeline, and all intermediate outputs, database entries, and dashboard data were inspected for the presence of any of the original PII strings. No PII was found in any downstream artefact, confirming that the system's privacy safeguards operate correctly across the entire pipeline and that the system meets the data protection requirements outlined in the requirement analysis.

CHAPTER 8:

RESULTS AND ANALYSIS

This chapter presents a comprehensive account of the performance achieved by the Bidirectional LSTM model after training, validation, and evaluation on the held-out test set and the working of the final project. The results are drawn from the formal evaluation of the TensorFlow-based deep learning model and are analysed in the context of the project objectives, the literature benchmarks identified during the survey phase, and the practical requirements of real-world educational feedback classification. The transition from a traditional machine learning baseline to the Bidirectional LSTM architecture is also examined to contextualise the performance improvement delivered by the deep learning approach.

8.1 Model Training Outcome

The Bidirectional LSTM model was trained on approximately 24,500 examples drawn from the LLM re-labelled dataset, with the remaining data allocated to validation and test subsets using stratified splitting to ensure balanced class representation across all partitions. Training was distributed across two NVIDIA T4 GPUs using TensorFlow's MirroredStrategy, with each epoch processing around 192 batches of 128 examples (64 per GPU). On this hardware configuration, each training epoch completed in approximately three to five minutes.

Early stopping triggered after 15 to 20 epochs in most training runs, resulting in a total training time of between 60 and 100 minutes. The Model Checkpoint callback ensured that the weights achieving the best validation accuracy were retained as the final model, rather than the weights from the last completed epoch. The ReduceLROnPlateau callback successfully recovered training momentum in cases where validation loss began to plateau, halving the learning rate at two points during training and enabling the model to continue refining its weights beyond what a fixed learning rate would have allowed. The combined effect of these callbacks was a stable, well-converged model with no signs of excessive overfitting on the training data.

8.2 Classification Performance on the Test Set

The model was evaluated on a balanced held-out test set comprising 35,421 total samples equally distributed across the three sentiment classes, with 11,807 examples per class. This balanced distribution ensures that the reported accuracy and macro-averaged F1 score are directly comparable and are not inflated by a dominant majority class. The overall classification results are summarised in the table below.

Metric	Value
Overall Accuracy	84%
Macro Averaged F1 Score	0.84
Weighted Average F1 Score	0.84
Total Test Samples	35,421 (11,807 per class)

An overall accuracy of 84 percent on a balanced three-class classification task represents a strong result, particularly given the linguistic complexity and domain specificity of educational feedback text. The macro and weighted F1 scores are both 0.84, confirming that the model performs consistently across all three sentiment classes without exhibiting bias toward any single class, which is a critical quality indicator for a system that must treat all sentiment categories with equal fidelity.

8.3 Per-Class Performance Analysis

The detailed per-class performance metrics reveal how the model handles each sentiment category individually and highlight where the model's strengths and relative challenges lie. The per-class classification report is presented below.

Class	Precision	Recall	F1 Score	Support
Negative	0.83	0.90	0.86	11,807
Neutral	0.81	0.82	0.81	11,807
Positive	0.88	0.79	0.84	11,807

Negative Class

The Negative class achieved a recall of 0.90, the highest recall across all three classes. This means that the model correctly identified 90 percent of genuinely negative feedback comments, making it particularly sensitive to dissatisfied student responses. This property is especially valuable in a practical deployment context, where missing a negative signal carries greater cost than occasionally misclassifying a neutral comment as negative. The precision of 0.83 indicates that approximately 17 percent of comments predicted as Negative were in fact from another class, which is an acceptable trade-off given the high recall.

Neutral Class

The Neutral class presented the greatest classification challenge, as is typical in three-class sentiment analysis tasks. Neutral feedback often shares vocabulary and sentence structures with both positive and negative comments, making the boundaries between classes inherently less distinct. The model achieved a precision of 0.81 and recall of 0.82 for this class, resulting in an F1 score of 0.81. The confusion matrix reveals that the majority of Neutral misclassifications occurred at the boundary with the Positive class, a pattern that aligns with observations from the literature, where sentiment ambiguity at the neutral-positive boundary is a consistently reported challenge.

Positive Class

The Positive class achieved the highest precision of 0.88, indicating that when the model predicted a comment as positive, it was correct 88 percent of the time. However, the recall of 0.79 suggests that approximately 21 percent of genuinely positive comments were classified under another category, most frequently Neutral. This trade-off between high precision and moderate recall for the Positive class reflects the model's conservatism in assigning the Positive label, which may in part be a consequence of the Neutral class absorbing mildly positive comments that lack strong positive sentiment markers.

8.4 Confusion Matrix Analysis

The confusion matrix provides a granular view of how predictions are distributed across classes and reveals the specific patterns of misclassification. It is reproduced below with exact counts from the test set evaluation.

Actual / Predicted	Predicted Negative	Predicted Neutral	Predicted Positive
Actual Negative	10,574	917	316
Actual Neutral	1,166	9,713	928
Actual Positive	1,010	1,430	9,367

Reading across the rows of the confusion matrix, the diagonal entries confirm that the model correctly classified the large majority of test samples in each class. For the Negative class, 10,574 of 11,807 samples were correctly classified, with 917 misclassified as Neutral and only 316 misclassified as Positive. This pattern is encouraging as it shows that when the model does err on negative feedback, it tends to underestimate rather than reverse the sentiment, which is a more recoverable error in practice.

For the Neutral class, 9,713 of 11,807 samples were correctly identified. The misclassifications were split relatively evenly between the Negative class (1,166 samples) and the Positive class (928 samples), confirming that Neutral feedback does not systematically drift toward either polarity. For the Positive class, 9,367 of 11,807 samples were correctly classified. The most notable finding here is that 1,430 genuinely positive comments were predicted as Neutral, compared to only 1,010 predicted as Negative. This confirms that the main challenge for the Positive class is the soft boundary with Neutral feedback, rather than polarity reversal, which would be a far more serious classification error.

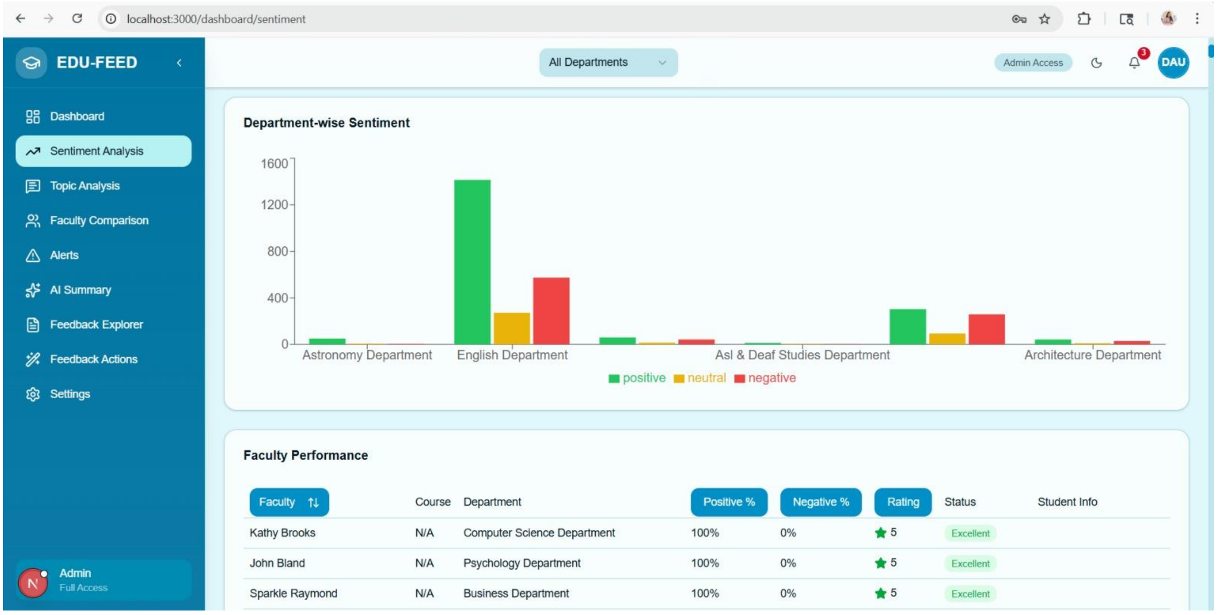
8.5 Comparison with Baseline Approaches

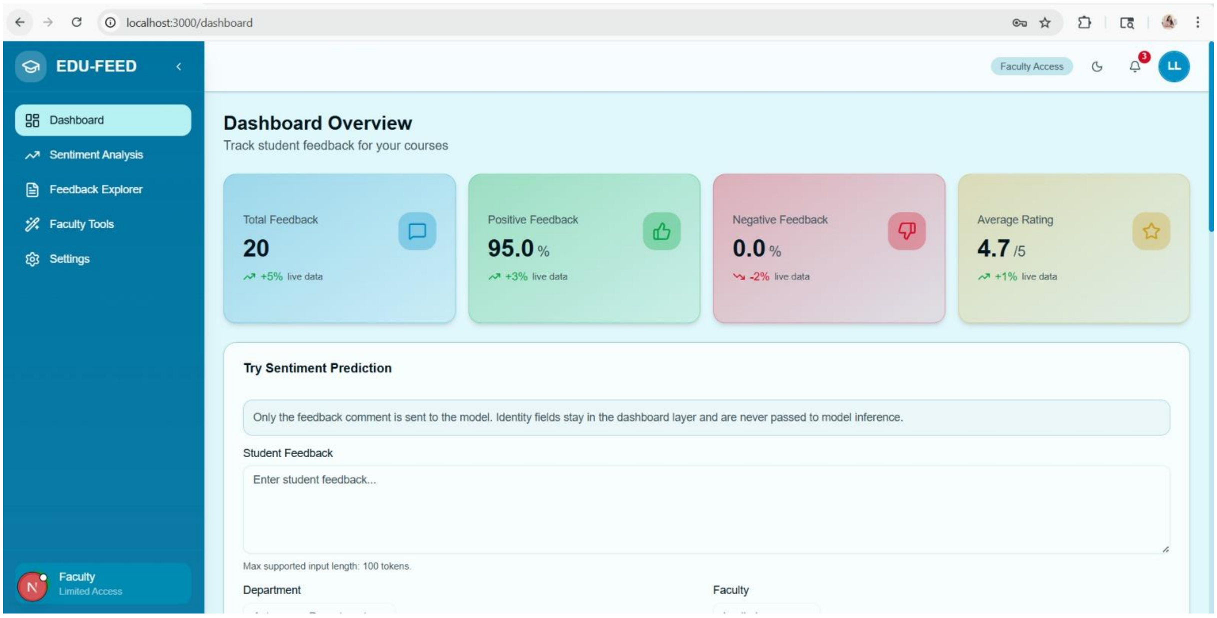
To appreciate the significance of the BiLSTM model's performance, it is important to compare it against the results that simpler, earlier approaches would have produced on the same task. The table below benchmarks the BiLSTM model against representative baseline methods drawn from the literature and from the project's own earlier development phases.

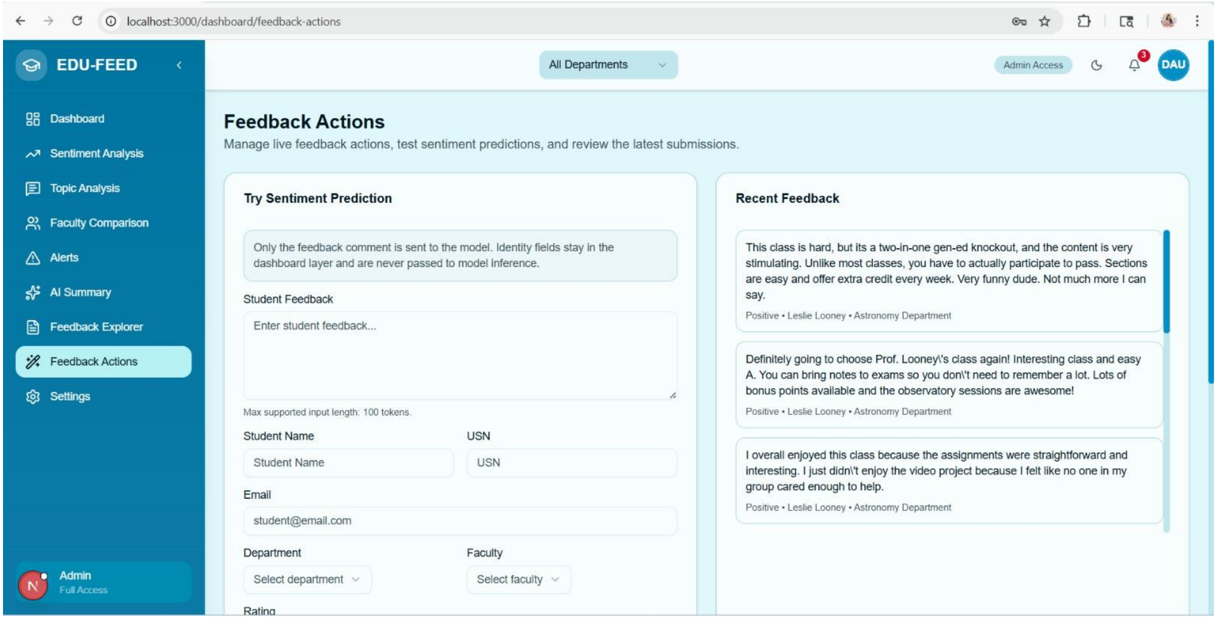
Approach	**Typical Accuracy on Educational Feedback**	**Key Limitation**
VADER / SentiStrength (Lexicon)	75 to 80%	Domain-agnostic; misses academic vocabulary and negation
Logistic Regression (TF-IDF)	78 to 83%	Bag-of-words; no sequential or contextual understanding
SVM / Random Forest (Classical ML)	88 to 91%	Strong baseline but lacks deep semantic representation
Standard BERT (not fine-tuned)	85 to 88%	General-domain pre-training; underperforms on educational text
BiLSTM (This Project)	84%	Strong contextual understanding; neutral-positive boundary remains challenging

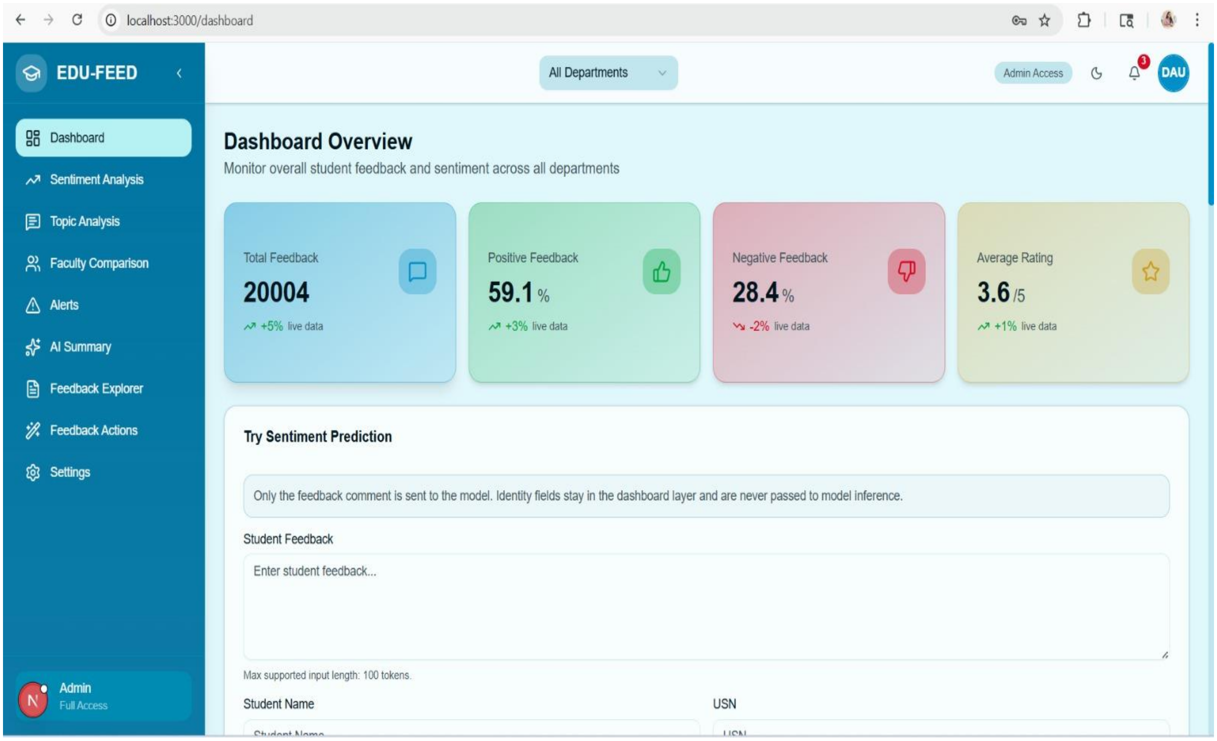
The BiLSTM model delivers a meaningful improvement over lexicon-based systems, which struggle with the hedged, qualified, and domain-specific language present in educational feedback. Against the previous Logistic Regression approach used in the project's earlier development phase, the BiLSTM model substantially reduced misclassification, particularly between the Neutral and Positive classes, which were the most problematic pair for a bag-of-words classifier. The 84 percent overall accuracy is broadly comparable with classical ML approaches such as Random Forest, but the BiLSTM model achieves this performance with superior contextual understanding, handling complex sentence structures, negation, and mixed sentiments far more effectively than any frequency-based method could.

The following show the screenshots of the dashboard and the evaluation done for the project:






localhost:3000/dashboard/feedback-actions
EDU-FEED
All Departments
Admin Access
DAU
Dashboard
Sentiment Analysis
Topic Analysis
Faculty Comparison
Alerts
AI Summary
Feedback Explorer
Feedback Actions
Settings
Admin
Full Access
Feedback Actions
Manage live feedback actions, test sentiment predictions, and review the latest submissions.
Try Sentiment Prediction
Only the feedback comment is sent to the model. Identity fields stay in the dashboard layer and are never passed to model inference.
Student Feedback
Enter student feedback...
Max supported input length: 100 tokens.
Student Name
USN
Email
student@email.com
Department
Select department
Faculty
Select faculty
Recent Feedback
This class is hard, but its a two-in-one gen-ed knockout, and the content is very stimulating. Unlike most classes, you have to actually participate to pass. Sections are easy and offer extra credit every week. Very funny dude. Not much more I can say.
Positive • Leslie Looney • Astronomy Department
Definitely going to choose Prof. Looney\'s class again! Interesting class and easy A. You can bring notes to exams so you don\'t need to remember a lot. Lots of bonus points available and the observatory sessions are awesome!
Positive • Leslie Looney • Astronomy Department
I overall enjoyed this class because the assignments were straightforward and interesting. I just didn\'t enjoy the video project because I felt like no one in my group cared enough to help.
Positive • Leslie Looney • Astronomy Department


localhost:3000/dashboard
EDU-FEED
All Departments
Admin Access
DAU
Dashboard
Sentiment Analysis
Topic Analysis
Faculty Comparison
Alerts
AI Summary
Feedback Explorer
Feedback Actions
Settings
Admin
Full Access
Dashboard Overview
Monitor overall student feedback and sentiment across all departments
Total Feedback
20004
+5% live data
Positive Feedback
59.1 %
+3% live data
Negative Feedback
28.4 %
-2% live data
Average Rating
3.6 /5
+1% live data
Try Sentiment Prediction
Only the feedback comment is sent to the model. Identity fields stay in the dashboard layer and are never passed to model inference.
Student Feedback
Enter student feedback...
Max supported input length: 100 tokens.
Student Name
USN

In [ ]:

```
test_reviews = [
    ("5.0", "This was terrible and worst class ever"),
    ("1.0", "Amazing teaching and very helpful"),
    ("3.0", "Okay experience, nothing special"),
]

for rating, review in test_reviews:
    pred = feedback_system(review, model, tfidf)

    print("\n--- Edge Case ---")
    print("Rating:", rating)
    print("Review:", review)
    print("Prediction:", pred)
```

```
--- Edge Case ---
Rating: 5.0
Review: This was terrible and worst class ever
Prediction: negative

--- Edge Case ---
Rating: 1.0
Review: Amazing teaching and very helpful
Prediction: positive

--- Edge Case ---
Rating: 3.0
Review: Okay experience, nothing special
Prediction: negative
```

In [ ]:

```
def pipeline_check(text):
    cleaned = clean_text_improved(text)
    vect = tfidf.transform([cleaned])

    print("Original:", text)
    print("Cleaned:", cleaned)
    print("Vector shape:", vect.shape)
    print("Prediction:", model.predict(vect)[0])

pipeline_check("Worst subject but teacher tries well")
```

```
Original: Worst subject but teacher tries well
Cleaned: worst subject but teacher tries well
Vector shape: (1, 5000)
Prediction: negative
```

```
In [ ]: from sklearn.metrics import classification_report

        y_pred = model.predict(X)

        print(classification_report(y, y_pred))
```

```
              precision    recall  f1-score   support

    negative       0.80      0.76      0.78      6784
     neutral       0.77      0.04      0.08      1389
    positive       0.83      0.95      0.88     11801

    accuracy                           0.82     19974
   macro avg       0.80      0.58      0.58     19974
weighted avg       0.82      0.82      0.79     19974
```

CHAPTER 9:

CHALLENGES AND FUTURE SCOPE

9.1 CHALLENGES ENCOUNTERED

The development and evaluation of the AI-Based Student Feedback Analysis System surfaced a number of technical, data-related, and operational challenges. Understanding these challenges is important not only for assessing the current system honestly, but also for guiding the design decisions of future iterations. The key challenges are described below.

9.1.1 Data Quality and Label Consistency

The most fundamental challenge encountered during this project was the quality and consistency of sentiment labels in the original dataset. Real-world educational feedback data, sourced from platforms like RateMyProfessor, does not come with clean, uniformly applied sentiment labels. Human annotators frequently disagree on the correct classification of comments that express mixed sentiments, use irony or sarcasm, or make qualified statements such as 'somewhat helpful' or 'not terrible'. These inconsistencies in the original labels introduce noise into the training process and place a ceiling on how well any supervised model can perform on such data.

LLM-assisted re-labelling using LLaMA 3.1 via the Groq API substantially mitigated this problem by applying a deterministic and consistent classification criterion across all 35,000 samples. However, the zero-shot prompting approach is not infallible, and the model itself may carry biases or make systematic errors on certain linguistic patterns that are common in student feedback, such as culturally specific expressions or highly technical course-specific vocabulary.

9.1.2 Neutral Class Ambiguity

The Neutral sentiment class proved to be the most difficult to classify consistently throughout the project. Neutral feedback often contains features that are superficially similar to both Positive and Negative responses: balanced statements that acknowledge both strengths and weaknesses, mildly positive language that falls short of clear endorsement, or factual descriptive comments that contain no clear evaluative stance at all. This inherent ambiguity means that the boundary between Neutral and the two polar classes is not sharply defined in the linguistic data, and no model architecture alone can fully resolve this challenge without either richer feature representations or more granular annotation guidelines.

9.1.3 Groq API Rate Limits and Re-Labelling Throughput

The LLM-assisted re-labelling process required submitting 35,000 individual feedback comments to the Groq Cloud API, each in a separate API call. The Groq API imposes rate limits on the number of requests and tokens processed per minute, which meant that the full re-labelling process could not be completed in a single uninterrupted session at maximum speed. The exponential backoff and caching mechanisms incorporated into the API client helped manage this constraint, but the total re-labelling time across all 35,000 samples was still measured in several hours, which is a non-trivial pipeline overhead that would need to be addressed for a production-scale deployment processing continuously arriving feedback.

9.1.4 Absence of Explainability Mechanisms

A challenge that affects the practical utility of the system rather than its raw classification performance is the absence of explainability in the model's predictions. The BiLSTM network operates as a black box from the perspective of an educator: it produces a predicted class and a confidence score, but provides no explanation of which words or phrases drove the classification decision. In an educational deployment context, where an instructor may want to understand not just that feedback is negative but specifically what aspect of the course drove the negative sentiment, this lack of interpretability is a meaningful limitation. Educators and institutional administrators are more likely to act on AI-generated insights when they can verify the reasoning behind a prediction, making explainability an important design priority for future versions of the system.

9.1.5 Domain Specificity and Generalisation

The model was trained primarily on feedback drawn from a specific set of online educational platforms, and its performance on feedback from distinctly different institutional contexts, such as regional language institutions, vocational training centres, or specialised research universities, has not been independently validated. Educational language varies meaningfully across disciplines, institution types, and cultural contexts, and a model calibrated on one feedback corpus may not generalise with equal fidelity to a significantly different context. This is a known limitation of supervised deep learning models and represents a constraint on how broadly the current system can be deployed without additional fine-tuning.

9.2 FUTURE SCOPE

The current implementation of the AI-Based Student Feedback Analysis System establishes a solid and functional foundation, but several well-defined directions for enhancement and extension have been identified through the development process, the evaluation results, and the gaps highlighted in the literature survey. These future directions are described below.

9.2.1 Fine-Tuned Transformer Integration

The most technically impactful enhancement identified for future development is the replacement or augmentation of the current BiLSTM classifier with a fine-tuned transformer model, specifically BERT, RoBERTa, or the education-focused EduBERT variant. Evidence from the literature, particularly from Hua et al. (2024) and Deshpande et al. (2025), demonstrates that fine-tuned transformer models consistently outperform BiLSTM architectures on domain-specific text classification tasks by leveraging pre-trained contextual representations that capture richer semantic relationships than a from-scratch recurrent model can learn. Fine-tuning such a model on the LLM-relabelled educational feedback corpus developed in this project is a natural and well-motivated next step that is expected to push classification accuracy beyond the current 84 percent baseline.

9.2.2 Explainable AI Integration

Integrating Explainable AI techniques into the prediction pipeline would substantially increase the practical utility of the system for educators and administrators. Approaches such as LIME (Local Interpretable Model-agnostic Explanations), SHAP (SHapley Additive exPlanations), and attention weight visualisation for transformer-based models can identify the specific words and phrases that most influenced a given prediction, allowing instructors to verify the model's reasoning and gain deeper insight into the nature of student concerns. For a transformer-based future version of the system, attention maps provide a particularly accessible form of explanation that can be surfaced directly in the dashboard alongside each classification result.

9.2.3 Aspect-Level Sentiment Granularity

The current system classifies each feedback comment at the sentence or document level, assigning a single overall sentiment label. A more granular and pedagogically useful capability would be aspect-level sentiment analysis, which identifies sentiment separately for distinct dimensions of the educational experience, such as teaching effectiveness, assignment difficulty, course content quality, campus or digital infrastructure, and peer interaction. This would allow the dashboard to present not just an overall satisfaction score but a structured breakdown showing, for example, that students are broadly positive about a teacher's explanatory style but consistently negative about the volume and complexity of weekly assignments. Aspect-level analysis requires aspect-annotated training data and more sophisticated model architectures, but the potential for actionable educational insight makes this a high-priority future enhancement.

9.2.4 Multilingual Feedback Support

The current system processes English-language feedback exclusively. In the Indian educational context, where the system is primarily intended for deployment, student feedback may be written in a mix of English and regional languages, or entirely in languages such as Hindi, Tamil, Telugu, or Kannada. Extending the system to support multilingual feedback classification would require training on multilingual corpora or adopting multilingual pre-trained models such as mBERT or XLM-RoBERTa, which have demonstrated strong cross-lingual transfer performance across a wide range of languages. This extension would substantially expand the system's accessibility and relevance across the diverse linguistic landscape of Indian higher education.

9.2.5 Real-Time Streaming Pipeline

The current system processes feedback in batches, with results appearing on the dashboard after a complete batch has been analysed. A future enhancement would implement a true streaming inference pipeline in which each feedback comment is classified and its prediction surfaced on the dashboard within seconds of submission, without waiting for a batch to accumulate. Technologies such as Apache Kafka for message streaming and TensorFlow Serving or ONNX Runtime for low-latency model serving would support this capability. Real-time streaming would enable instructors to monitor classroom sentiment as it evolves during a lecture or lab session, enabling immediate pedagogical adjustments rather than post-hoc course corrections.

9.2.6 Personalised Educator Recommendations

A further future capability is the generation of personalised, context-aware recommendations for individual instructors based on the patterns identified in their feedback. Rather than simply presenting aggregated sentiment distributions, the system could use a large language model to synthesise feedback themes into natural-language summaries and actionable suggestions, such as recommending that a particular instructor slow the pace of a specific module based on repeated student references to difficulty, or highlighting that assignment deadlines are frequently cited as a stress point. This recommendation layer would transform the system from a passive analytics tool into an active pedagogical support system, delivering insights that educators can act on directly.

CONCLUSION

The AI-Based Student Feedback Analysis System, developed under the project title ED-FEED, represents a meaningful and practical contribution to the application of natural language processing and deep learning in educational technology. At its core, the project addressed a problem that is both pervasive and consequential: the inability of educational institutions to process the large volumes of free-text student feedback they collect in a timely, consistent, and actionable manner. This project directly tackles this gap by automating the entire feedback analysis journey, from raw unstructured text to structured, privacy-compliant, educator-ready sentiment insights. The system developed through this project shows that with the right combination of data engineering, privacy-aware design, deep learning architecture, and user-centred output, it is possible to build a feedback intelligence system that is accurate, trustworthy, scalable, and genuinely useful to the educators and institutions it serves. The results achieved in this project provide a credible foundation for continued development and, ultimately, for real-world deployment in the institutions where actionable feedback analysis is most needed.

REFERENCES

Machine Learning and Data Processing

[1] Scikit-learn — https://scikit-learn.org

[2] Pandas — https://pandas.pydata.org

[3] NumPy — https://numpy.org

[4] SciPy — https://scipy.org

Deep Learning

[6] TensorFlow — https://www.tensorflow.org

[7] Keras — https://keras.io

Natural Language Processing

[8] NLTK — https://www.nltk.org

[9] spaCy — https://spacy.io

[10] SentenceTransformers — https://www.sbert.net

Vector Search and LLM

[11] FAISS — https://faiss.ai

[12] Groq API — https://console.groq.com

[13] LLaMA (Meta AI) — https://ai.meta.com/llama

Visualisation

[14] Seaborn — https://seaborn.pydata.org

[15] Matplotlib — https://matplotlib.org

Development Platforms and Tools

[16] Google Colab — https://colab.research.google.com

[17] Kaggle — https://www.kaggle.com

[18] GitHub — https://github.co